Lec-13

① SJF (shortest Job first) (Non-preemptive)

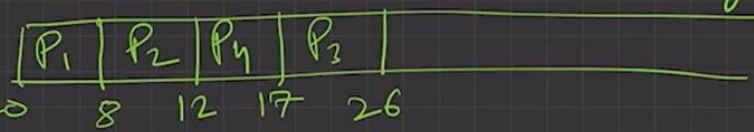
– Process with least BT will get CPU.



P	AT	BT	CT	TAT	WT
P_1	0	8	8	8	0
P_2	1	4	12	11	7
P_3	2	9	26	24	15
P_4	3	5	17	14	9

Avg WT 7.75s

Non-Pr.

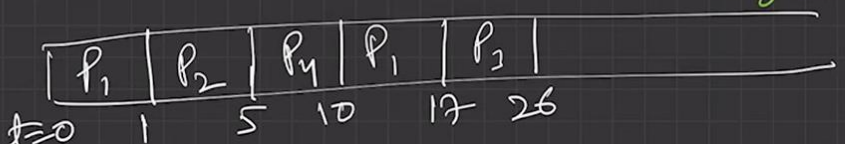


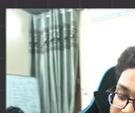
Criteria! AT + BT

* Preemptive SJF :- Leetstock

P	AT	BT	CT	TAT	WT
P_1	0	~~8~~ 7	17	17	9
P_2	1	4	5	4	0
P_3	2	9	26	24	15
P_4	3	5	10	7	2

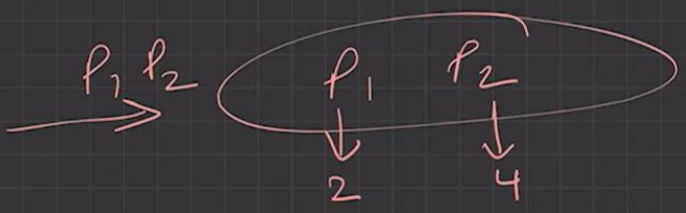
Avg WT :⇒ 6.5 units





* Priority scheduling :-

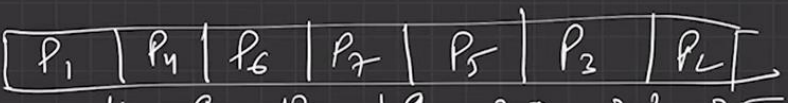
* ⟶ Assign Priority to each Job



① Non-Preemptive :-

P	Priority	AT	BT	CT	TAT	WT
1	2	0	4	4	4	0
2	4	1	2	25	24	22
3	6	2	3	23	22	19
4	10	3	5	9	6	1
5	8	4	1	20	16	15
6	12	5	4	13	8	4
7	9	6	6	19	13	7

Avg. 9.714



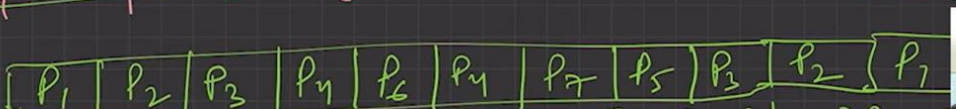


* Preemptive Priority scheduling,

Pno.	P*	AT	BT	CT
1	2	0	~~4~~ 3	25
2	4	1	~~2~~ 1	22
3	6	2	~~3~~ 2	21
4	10	3	~~5~~ 2	12
5	8	4	1	19
6	12	5	4	9
7	9	6	6	18

Find W Avg WT

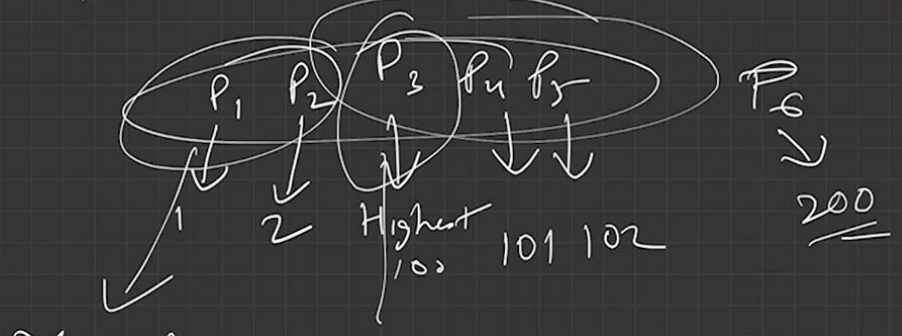
Avg = 11.4



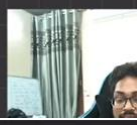


Biggest Drawback ⇒ Non Pr Preemb

Indefinite waiting or Extreme Starvation



X CPU Term



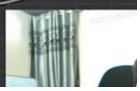
Solution to indefinite waiting

Ageing :→

Gradually Ting Priorities of Lowest Jobs

15 min → Lowest Priorities Jobs

Priority +1



* Round-Robin (RR)
- Most Popular
- FCFS (Preemptive) version
- Criteria: AT + TQ, BTX
- design Time-Sharing
- Easy to implent.

P	AT	BT
1	0	4
2	1	5
3	2	2
4	3	1
5	4	6
6	6	3

TQ = 2s

TQ = 1s

Queue:



